DICE, EXPECTATION, CONDITIONAL PROBABILITY AND BLUFF ON ONE PAGE

Perudo: probability under a cup

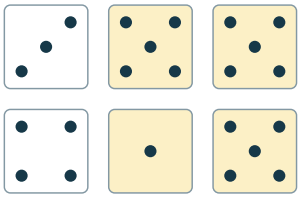
Your own dice are evidence. The hidden dice are uncertainty. A bid turns both into a claim.

1 / SEE YOUR DICE, REASON ABOUT EVERYBODY'S

A bid means "at least", and ones are wild

Roll and conceal the dice; inspect only your own. Take turns raising a bid or calling **Dudo** ("I doubt"). Reveal: if the claim fails, the bidder loses a die; if it holds, the challenger loses one. Re-roll; the last player with dice wins. For ordinary bids, raise the quantity, or keep it and raise the face.

Classroom example: six players with six dice each, so 36 dice in all. You see the hand below. A bid of "fourteen fives" needs at least 14 dice showing either 5 or a wild 1, across *all* cups.



Four known matches: three fives and one ace.

There are $36-6=30$ hidden dice. Each supports a five-bid with probability $p=\frac{1}{6}+\frac{1}{6}=\frac{1}{3}$. Let X count those hidden matches.

$$T=4+X, \qquad \mathbb{E}(X)=30\times\tfrac{1}{3}=10.$$

$$\mathbb{E}(T \mid \text{your hand}) = 4+10 = 14$$

Without wild ones, the same hand gives $3+30/6=8$ expected fives.

Before seeing any dice, the wild-one expectation was $36/3=12$. Your unusually helpful hand raises it to 14; **the mean is not a guarantee**.

2 / TURN THE CLAIM INTO A PROBABILITY

Count successes: the binomial model

For n independent hidden dice, each with match probability p,

$$X\sim \mathrm{Bin}(n,p), \qquad \mathbb{E}(X)=np.$$

Exactly j matches can occupy $\binom{n}{j}$ sets of positions. Each arrangement has probability $p^j(1-p)^{n-j}$, so

$$\mathbb{P}(X=j)=\binom{n}{j}p^j(1-p)^{n-j}.$$

In this hand, "fourteen fives" needs $14-4=10$ or more hidden matches:

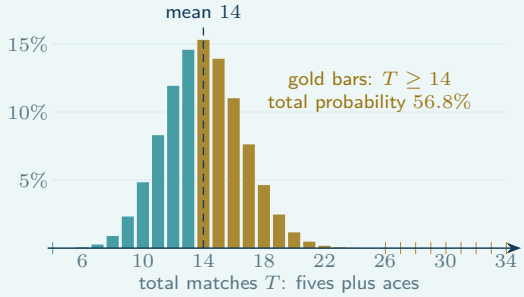
$$\mathbb{P}(T\geq 14 \mid \text{hand})=\sum_{j=10}^{30}\binom{30}{j}\left(\tfrac{1}{3}\right)^j\left(\tfrac{2}{3}\right)^{30-j}$$

$$\mathbb{P}(T \geq 14 \mid \text{hand}) \approx 56.8\%$$

Exactly 14 has probability only 15.3%. Perudo asks for a **tail probability**: add every outcome that makes the bid true.

3 / READ THE CHANCES FROM THE DISTRIBUTION

Higher bids leave less probability



Bid	Hidden needed	Chance true
12 fives	≥ 8	83.3%
14 fives	≥ 10	56.8%
16 fives	≥ 12	27.6%
18 fives	≥ 14	9.0%

4 / SEPARATE DICE EVIDENCE FROM PLAYERS' BEHAVIOUR

Independence, information and bluff

Seeing three fives in your hand does **not** make a hidden five less likely. Independent dice do not use up a fixed supply of each face. Only your known contribution has changed.

But another player's bid may reveal information about their hand. Updating on bids needs a model of how that player bids or bluffs; the binomial calculation above uses no such information.

If the chance a bid is true is r, calling Dudo loses you a die with probability r. Choosing between challenging and raising also depends on what happens next; 50% is no universal strategy rule.

5 / MAKE A PREDICTION, THEN TEST IT

Keep the hand fixed; re-roll the unknowns

Keep the six displayed dice unchanged. Re-roll the other 30 dice 50 times. Each time, record $T=4+X$, then compare:

$$\text{mean of } T \text{ with } 14; \qquad \frac{\#\{T\geq 14\}}{50} \text{ with } 0.568.$$

Expect roughly 28 successful claims in 50 trials, with random variation. Re-rolling all 36 dice would instead test the unconditional model, whose mean is 12.

Generalise. If s visible dice support the bid and n are hidden, then $T=s+X$, $X\sim\mathrm{Bin}(n,p)$:

$$\mathbb{E}(T \mid \text{hand})=s+np, \qquad \mathbb{P}(T\geq q)=\mathbb{P}(X\geq q-s).$$

Conventions. Standard Perudo starts with five dice per player; six each is the classroom variant here. Fair, independent dice; ordinary non-ace bids with wild ones. An ace-only bid has $p=1/6$. Special bidding and Palifico rules are outside this sheet's scope.

Rules: The official home of Perudo; Asmodee rulebook. Probabilities are calculated for the stated hand and assumptions.

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